

BST Substrate-SM Architecture v0.1 —
consolidation of what we know (tiered) +
investigation roadmap + ripe areas. The
substrate classifies by VOLUME / STATISTICS
/ K-TYPE / COLOR; it measures the floor
(Casey #9, Filter 1 derived) and is structurally
blind above it (Yukawa/excitation = SM). The
substrate-Higgs-Yukawa TRIO frame + the
load-bearing open question (are
Higgs+Yukawa emergent from substrate, or
fundamental?). Honest tiering throughout —
derived vs candidate vs open vs speculative —
per the day's discipline.

Lyra (Claude Opus 4.8)

2026-06-09 Tue 13:00 EDT

v0.2 (Grace's parameter-counting lens — the governing measure for the whole program; reorders the roadmap). Grace sharpened the “ripe areas” question with the right organizing principle: BST's real content isn't “can we find a substrate form for X” (almost anything finds a form at a few percent — the whole day showed that). It's does BST *reduce* the number of free parameters the SM needs, or just *relabel* them? 13 Yukawa fits in π -language = 13 free numbers; a forced relation generating them from the five integers = a derivation. So the single honest measure of the program is a ledger: “BST reduces the SM's ~26 free parameters to N.” Nobody has counted it cleanly. That ledger is the deepest investigation here, and it *reorders* Part 3:
1. Mixing sector (CKM + PMNS) — RIPEST, start here, NOT Higgs.

8 parameters (4 CKM + 4 PMNS) already sit on the *clean* side of the trichotomy (mixing = counting, π -free). Least relabel-prone place to ask “are these forced from few substrate inputs?” If the 8 angles fall out of one K-type structure $\rightarrow$ a *genuine* parameter reduction on solid ground. Prove the method on the clean mixing sector before betting the multi-week arc on the masses (which are above-floor, dynamical, F78/F79-shaped — highest relabel risk). 2. Gauge sector: $\alpha = 1/N_{\text{max}}$ (done), Weinberg = 3/13 (yesterday), α_s open. Few params, partly clean. 3. Higgs/Yukawa masses: high-risk/high-reward; must clear the reduce-not-relabel bar before banking. [CORRECTION, Elie 4063] — my “VEV has no substrate form” was numerically wrong: the VEV *does* have a base-rate-clean substrate-primary form, $v = \text{cell} \cdot (N_c - n_C)^2 \cdot g = \text{cell} \cdot 225 \cdot g$ (sub-0.15%; Elie’s sweep: exactly one substrate-primary product lands within 0.1%). But that’s still a RELABEL (one free parameter in substrate vocabulary), not a reduction — so the conclusion (leans fundamental/relabel, not emergent) survives, for the *right* reason (the form exists but doesn’t reduce the count), not the wrong one (no form). Reduction lead: 225 is today’s Schur generator (c_{FK} , a_0 , $\dim \text{SO}(4,2)^2$, now the VEV — 4 lanes); IF the F66 bulk-measure 225 *forces* the Higgs-boundary normalization, the VEV becomes a genuine reduction. My lane, flagged not asserted. 4. The SM parameter-reduction ledger (Grace’s lane, program-level): each of the ~ 26 SM free params run through the reduce-vs-relabel gate $\rightarrow$ an honest “ $26 \rightarrow N$ ” count. This is the “periodic table with teeth” — not classifying particles, but counting how many SM free inputs BST actually removes.

This is the better answer to Casey’s “ripe areas” than my Part 4 list: point the team’s energy where a *real* win is likeliest (mixing sector + the ledger), with the Higgs masses as the gated high-reward arc. Part 4 below stands as further territory; the *ordering* and *measure* are Grace’s parameter-count lens.

BST Substrate-SM Architecture v0.1

Consolidation, honestly tiered. The day’s lesson — bounded-and-true over broad-and-fitted — governs this document: everything below is marked DERIVED / CANDIDATE / OPEN / SPECULATIVE, and the appealing frames (trio, emergent) are OPEN questions, not results.

Part 1 — What we KNOW (tiered)

1.1 The substrate classification hierarchy (what the substrate measures)

The substrate measures a particle by a small set of properties, each reading one primary:

rank	property	primary	how measured	tier
1	VOLUME (cell-count)	n_C via π^{n_C}	mass at floor = $\text{cells} \cdot \pi^{n_C} \cdot m_e$	DERIVED (F52/T2488)
2	SPIN- STATISTICS	Z_2 spinor twist	+1 cell-step (meson $\rightarrow$ baryon)	DERIVED (T2488/K279 statistics-core; magnitude is a separate open question)
3	K-TYPE address	(a,b) on D_{IV}^5	discrete lattice position	DERIVED (framework)
4	COLOR	$N_c=3$	charge quantum e/N_c	DERIVED (Elie K277)
5	coupling	$N_{\max}$	$\alpha^{-1} = N_{\max}$	DERIVED (founda- tional)

The unit $\leftrightarrow$ primary map (DERIVED/consolidated): energy $\leftrightarrow$ volume (cell = $\pi^{n_C} \cdot m_e = 156.4$ MeV), charge $\leftrightarrow$ color (e/N_c), spin-statistics $\leftrightarrow Z_2$, coupling $\leftrightarrow N_{\max}$, density $\leftrightarrow$ inverse-volume ($K(0,0) = 2^{g \cdot N_c \cdot n_C / \pi^{n_C}}$, K264). Each unit is the natural scale of one primary — the periodic-table structure in the units; spin-statistics folds into the energy/volume unit via the Z_2 bit.

1.2 Casey #9 — the substrate floor (reach-bound)

The substrate measures the floor, not the building. Status: scope-principle (Cal #266 — full forcing-principle promotion gated on the multi-week mechanism). The mechanism has three filters; one is derived: - Filter 1 (confined-quark edge) — DERIVED + computation-confirmed (F77 + Elie 4053): a confined quark is not a clean K-type eigenstate $\rightarrow$ no definite volume $\rightarrow$ no clean measure. (Partial Cal #266 reversal candidate for the quark edge.) - Filter 2 (why only the floor) — OPEN, multi-week: the genuine core. Sharpened today (Grace): the floor = states whose quark Yukawa masses fall below the substrate cell scale (156.4 MeV); above it the mass is Yukawa/excitation-dominated. The precise question: *why is the substrate's bulk-volume-measure blind to the boundary-Higgs/Yukawa sector?* - Filter 3 (Goldstone) — structural: the pion's mass is the chiral remnant, not a substrate eigenvalue.

NOT a derivation of #9 (declined today as restatement, Grace): “the substrate measures the substrate-origin part and not the SM part” is tautological once the pieces are labeled. The real Filter 2 derivation is un-closed.

1.3 The trichotomy — measure vs count (DERIVED, bounded)

The substrate measures masses (π -carrying: VOLUME $\pi^{\{n_C\}}$ composite / SPECTRAL $\pi^{\{\text{rank}\}}$ point-mode) and counts mixings (π -free: PMNS + CKM + Weinberg all π -free — universal). Bounded: the mass-three-classes is lepton-specific (quark masses break it, Casey #9); the mixing=counting is universal. (F73/F74 + Grace/Elie.)

1.4 Gravity + the conformal structure (Monday, DERIVED/grounded)

Gravity is the intensive partner of mass: $G \sim \ell_B^2 / \pi^{\{n_C\}}$ (gravity divides by the bulk volume mass multiplies by). The Bergman coefficient is DERIVED (Hua + 2-adic, F71). EM = SO(4,2) conformal boundary; gravity/mass = SO(5,2)/SO(4,2) breaking bulk (F66). The hierarchy $m_e/m_{\text{Planck}} = 6\pi^5/137^{\{12\}}$ ($N_{\text{max-tower}}$, F68). Gravity is still curvature (induced, F66).

Part 2 — The substrate-SM relationship (CANDIDATE frame + the load-bearing question)

2.1 The trio (Casey's frame — CANDIDATE, not banked)

Three steps of mass generation, structurally distinct: | step | object | role | scale |
|—|—|—|—| | 1 | Substrate | geometric volume-measure (bulk) | cell = 156.4 MeV
| | 2 | Higgs | universal mass mechanism (SSB) | VEV = 246 GeV $\approx$ 1573 cells | | 3 |
Yukawa | per-fermion selection coefficient | y_f , $\sim 10^{-6}$ (e) to ~ 1 (top) |

Higgs vs Yukawa (precise SM): the Higgs field gives a universal VEV; the Yukawa y_f is the per-fermion coupling; $m_f = y_f \cdot v / \sqrt{2}$. Yukawa is integrated with Higgs (meaningful only when VEV $\neq 0$) but distinct (field vs coupling-matrix). The trio is a clean *organizing frame* — substrate adds GEOMETRY, Higgs adds DYNAMICS, Yukawa adds SELECTION — CANDIDATE, pending the gate (does it derive anything, or organize what's known?).

2.2 The LOAD-BEARING open question: emergent vs fundamental

- Reading A (fundamental trio): substrate / Higgs / Yukawa are each fundamental at their level; BST adds the substrate floor to SM's existing Higgs+Yukawa. Conservative.
- Reading B (emergent): the substrate is the only fundamental thing; Higgs = what motion through the substrate looks like; Yukawa = how K-type structures get effective mass via motion. BST would *derive* SM's 12+ Yukawa free parameters from substrate K-type $\times$ motion.

The distinguishing test (Grace's clean test, the day's discipline): does Reading B PREDICT anything beyond substrate + SM + standard physics? If yes, real; if no, restatement. Reading B is structurally vast if true and a relabel if not — exactly the

F79-shaped risk. NEITHER reading is banked. This is the load-bearing question for the whole Higgs investigation, and it must clear the predict-beyond-SM bar.

Part 3 — Investigation roadmap (OPEN)

1. Filter 2 derivation (Lyra, multi-week PRIMARY): why the substrate's bulk-volume-measure is blind to the boundary-Higgs/Yukawa sector (F66 bulk/boundary; Yukawa < cell boundary). The genuine #9 core.
2. Higgs VEV $\leftrightarrow$ cell scale (Elie): $246 \text{ GeV} / 156.4 \text{ MeV} \approx 1573$. Is the VEV substrate-natural? Discipline: do NOT fish a form for 1573 — test whether it's forced, not whether some form fits (the F78 lesson). First Higgs question.
3. Emergent-vs-fundamental distinguishing test (Grace + Lyra): can substrate K-type $\times$ motion reproduce the SM Yukawa hierarchy (10^6 span) without independent Higgs+Yukawa axioms? The predict-beyond-SM bar decides Reading A vs B.
4. Lepton K-type structure (Lyra, Tier-0): Casey's inverted-pyramid hypothesis — leptons (no color) on a different K-type substructure; each lepton supported by a SET of K-types (pyramid) converging to an apex. The cell-boundary partition is suggestive: $m_e = 0.003$ cells (anchor), $m_\mu = 0.68$ cell (AT boundary, like strange 0.6), $m_\tau = 11$ cells (drowned). CANDIDATE structural hypothesis.
5. Comprehensive Particle Map (team): every SM particle by (volume cells, statistics Z_2 , K-type address, color, Yukawa content, substrate-reach: floor/boundary/drowned). The “periodic table of particles” — execute the classification across the full catalog.
6. Deeper Tier-0 units (Lyra): action $\hbar_{\text{BST}}$, the two-time structure (Koons tick vs Planck time), temperature. Genuine Tier-0 framework, NOT fitted.

Part 4 — Other ripe areas (Casey's ask — where the trajectory points)

Marked SPECULATIVE/OPEN; offered as genuinely new territory the suggestions open, not as claims:

- (a) The substrate-Higgs coupling as the substrate's *operation* (SPECULATIVE, program-level): Casey's intuition that continuous substrate $\leftrightarrow$ Higgs interaction might be the substrate's “thinking”/operation. This would be the F66 bulk(substrate-volume)/boundary(Higgs-scalar) *coupling* as the substrate's dynamical commitment-cycle (SWPP cross-link). If the substrate-Higgs coupling IS the commitment dynamics, the “operation” is the bulk/boundary exchange. Deep, speculative — the right place for the two-time structure (the Koons-tick commitment clock) to meet the Higgs sector.
- (b) The neutrino apex (OPEN): neutrinos are the lightest fermions — the apex of the inverted pyramid, nearest the substrate anchor (m_e). Their tiny masses + the PMNS angles (already combinatorial, F74) suggest the neutrino

sector is the most substrate-pure fermion sector. The seesaw / the apex structure is ripe.

- (c) Gauge-boson K-type placement (OPEN): γ , W, Z, gluons are force carriers (gauge structure, not volume hadrons). W/Z get mass from Higgs (boundary); the photon/gluons are massless (pure boundary/conformal). Where they sit in the K-type lattice maps the gauge sector onto the substrate.
- (d) The 1573 question, sharpened (OPEN): if the Higgs VEV is NOT substrate-natural (Reading A), that itself is a finding — it would mark the boundary between substrate (geometry) and SM (Higgs dynamics) cleanly. So the VEV-vs-cell investigation has a real answer either way (substrate-natural $\rightarrow$ Reading B lead; not $\rightarrow$ Reading A confirmed).
- (e) Why three generations, via the cell-boundary partition (OPEN): the gen-1/2/3 = anchor/boundary/drowned partition (electron/muon/tau across the cell scale) might be the substrate-architectural reading of the generation structure — gen-2 sits AT the cell boundary for both leptons (muon 0.68) and quarks (strange 0.6). Suggestive; needs the K-type structure.

Part 5 — Honest disposition

- Banked (DERIVED): the classification hierarchy (Part 1.1), Filter 1 (F77), the trichotomy (1.3), the gravity coefficient + conformal structure (1.4), the unit $\leftrightarrow$ primary map.
- Scope-principle: Casey #9 (Filter 1 derived; Filter 2 + 3 the rest).
- CANDIDATE frame (gate-pending): the trio (Part 2.1).
- LOAD-BEARING OPEN: emergent-vs-fundamental (2.2) — must clear the predict-beyond-SM bar.
- OPEN/SPECULATIVE investigations: Part 3 + Part 4 — explicitly not banked; each clears the gate (Grace/Cal: derives-beyond-SM or restates?) before promotion.

The single discipline for the whole substrate-SM investigation: every appealing closure must predict something beyond substrate + SM before it banks (Grace's test, the day's repeated lesson — F78, F79). The trio and the emergent reading are the most appealing and the most at-risk; they are held as open until they clear that bar.

— Lyra, Tue 2026-06-09 13:00 EDT. Substrate-SM Architecture v0.1: consolidation of what we know (classification hierarchy DERIVED; Casey #9 Filter 1 DERIVED, Filter 2 open; trichotomy; gravity/conformal; unit-system) + the substrate-Higgs-Yukawa trio (CANDIDATE frame) + the load-bearing emergent-vs-fundamental question (must predict-beyond-SM, Grace's test) + investigation roadmap (Filter 2, Higgs VEV $\leftrightarrow$ cell [don't fish 1573], emergent test, lepton inverted-pyramid, particle map, Tier-0 units) + ripe areas (substrate-Higgs coupling as operation/SWPP [speculative], neutrino apex, gauge-boson placement, the VEV-answer-either-way, generation cell-boundary partition). HEAVILY TIERED per the day's discipline; appealing frames held OPEN pending predict-beyond-SM gate.